

Лабораторная работа №2

Структуры данных

Владимир Базлов

22 сентября 2026

Российский университет дружбы народов, Москва, Россия

```
[3]: t0 = ()  
favouritelang = ("Pythone", "Julia", "R")  
x1 = (1, 2, 3)  
x2 = (1, 2.0, "tmp")  
x3 = (a=2, b=1+2)  
  
length(x2)  
x2[1], x2[2], x2[3]  
c = x1[2] + x1[3]  
x3.a, x3.b, x3[2]  
in("tmp", x2), 0 in x2
```

```
[3]: (true, false)
```

```
[5]: phonebook = Dict("Иванов И.И." => ("867-5309", "333-5544"), "Бухгалтерия"  
  
      keys(phonebook)  
      values(phonebook)  
      pairs(phonebook)  
      haskey(phonebook, "Иванов И.И.")  
  
      phonebook["Сидоров П.С."] = "555-3344"  
      pop!(phonebook, "Иванов И.И.")  
  
      a = Dict("foo" => 0.0, "bar" => 42.0)  
      b = Dict("baz" => 17, "bar" => 13.0)  
      merge(a, b)  
      merge(b, a)
```

```
[5]: Dict{String, Real} with 3 entries:  
      "bar" => 42.0  
      "baz" => 17  
      "foo" => 0.0
```

```
[7]: A = Set([1, 3, 4, 5])  
     B = Set("abrakadabra")  
  
     S1 = Set([1,2]); S2 = Set([3,4])  
     issetequal(S1, S2)  
  
     S3 = Set([1,2,2,3,1,2,3,2,1])  
     S4 = Set([2,3,1])  
     issetequal(S3, S4)  
  
     C = union(S1, S2)  
     D = intersect(S1, S3)  
     E = setdiff(S3, S1)  
     issetequal(S1, S4)  
  
     push!(S4, 99)  
     pop!(S4)
```

[7]: 2

```
[11]: empty_array_1 = []  
      empty_array_2 = (Int64)[]  
      empty_array_3 = (Float64)[]  
  
      a = [1, 2, 3]  
      b = [1 2 3]  
  
      A = [[1, 2, 3] [4, 5, 6] [7, 8, 9]]  
      B = [[1 2 3] [4 5 6] [7 8 9]]  
  
      c = rand(1, 8)  
      C = rand(2, 3)  
      D = rand(4, 3, 2)  
  
      roots = [sqrt(i) for i in 1:10]
```

```
[11]: 10-element Vector{Float64}:  
       1.0  
       1.4142135623730951  
       1.7320508075688772  
       2.0  
       2.23606797749979  
       2.449489742783178  
       2.6457513110645907  
       2.8284271247461903  
       3.0  
       3.1622776601683795
```

```
[13]: isempty(t0)
      length(x1)
      in(2, x1)
      unique([1,2,2,3,1])
      reduce(+, [1,2,3,4])
      maximum([1,5,3])
      minimum([1,5,3])
```

```
[13]: 1
```

```
[15]: x = rand(1:1000, 50)  
      y = rand(1:1000, 50)
```



```
[15]: 50-element Vector{Int64}:  
      235  
      972  
      819  
      809  
      880  
      911  
      723  
      770  
      723  
      904  
      350  
      712  
      602  
      ⋮  
      834  
      310  
      394  
      129  
      776  
      857  
      487  
      398  
      924  
      416  
      979  
      887
```

```
[43]: squares = [i^2 for i in 1:100]  
      length(squares)  
      squares[1]  
      squares[100]
```

```
[43]: 10000
```

```
[45]: using Pkg  
      Pkg.add("Primes")
```

```
Updating registry at `C:\Users\Asus\.julia\registries\General.toml`  
Resolving package versions...  
Installed Primes ————— v0.5.7  
Installed IntegerMathUtils - v0.1.4  
Updating `C:\Users\Asus\.julia\environments\v1.12\Project.toml`  
[27ebfcd6] + Primes v0.5.7  
Updating `C:\Users\Asus\.julia\environments\v1.12\Manifest.toml`  
[18e54dd8] + IntegerMathUtils v0.1.4  
[27ebfcd6] + Primes v0.5.7  
Precompiling packages...  
 1392.5 ms ✓ IntegerMathUtils  
   771.3 ms ✓ Primes  
2 dependencies successfully precompiled in 5 seconds. 348 already pre  
compiled.
```

```
[47]: using Primes
```

```
primes_arr = primes(1, 1000)[1:168]  
slice = primes_arr[89:99]
```

```
[47]: 11-element Vector{Int64}:
```

```
461  
463  
467  
479  
487  
491  
499  
503  
509  
521  
523
```

```
[49]: sum(i^3 + 4i^2 for i in 10:100)
```



```
[49]: 26852735
```

```
[51]: M = 25  
      sum((2i)/i + (3i)/(i^2) for i in 1:M)
```

```
[51]: 61.44787453326052
```

```
[53]: term = 1.0
      total = 1.0
      for k in 2:2:38
          term *= k / (k + 1)
          total += term
      end
      total
```

```
[53]: 6.976346137897618
```

Итоги работы

В ходе лабораторной работы:

1. Изучены структуры данных, реализованные в Julia: кортежи, словари, множества, массивы.
2. Освоены операции над ними: индексация, доступ по ключу, теоретико-множественные операции, работа с многомерными массивами.
3. Изучены общие функции `isempty()`, `length()`, `in()`, `unique()`, `reduce()`, `maximum()`, `minimum()`.
4. Получены навыки работы с пакетами Julia и оформления результатов в Jupyter Lab.